EEE554 – Probability and Random Processes

Module 3: Conditional Probability and Bayes' Rule

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From Probability to Conditional Probability

So far, probabilities have been computed using a probability space

$$(\Omega, \mathcal{F}, P).$$

For an event

$$A \in \mathcal{F},$$

we compute

$$P(A).$$

This represents our probability of A given the information currently available. But what happens when we acquire new information?

Why Conditional Probability?

In many applications, we accumulate *additional information*:

- Probability of packet loss given network congestion.
- Probability of a communication error given the transmitted symbol.
- Probability of disease given a positive medical test.
- Probability of target detection given the received signal.

Conditioning incorporates the available information into the probability calculation.

Conditional Probability

Suppose we learn that an event B has occurred.

Outcomes outside B are no longer possible.

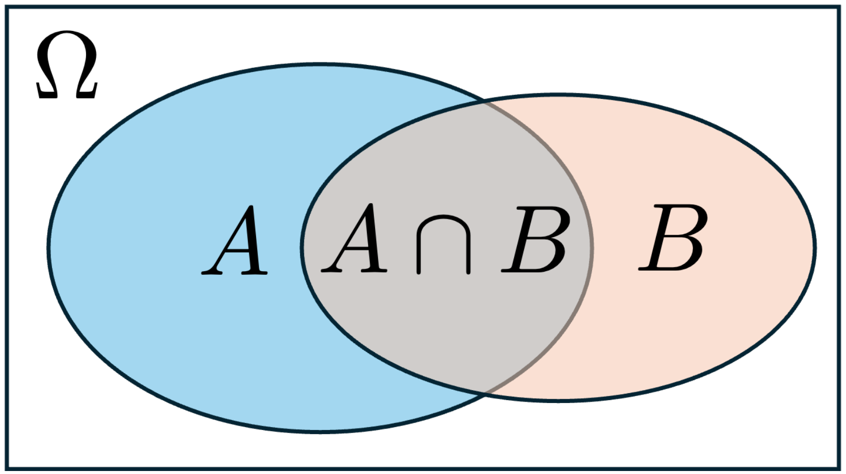
Question:

$\boxed{\text{How should this new information change the probability of } A\text{?}}$

Knowing that B occurred restricts our attention to B.

Within this restricted sample space, the outcomes for which A also occurs are

$$A \cap B.$$



Therefore, we renormalize probability within B:

$$\boxed{P(A|B) = \frac{P(A \cap B)}{P(B)}.}$$

This naturally leads to the following definition.

Definition 1. *For* $A, B \in \mathcal{F}$ *with*

$$P(B) > 0,$$

the conditional probability of A *given* B *is*

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

Properties

If $A \subseteq B$, then

$$P(A|B) = \frac{P(A)}{P(B)} \geq P(A).$$

Knowing that B occurred makes A more likely.

If $B \subseteq A$, then

$$P(A|B) = 1.$$

Once B occurs, A is certain.

If $A \cap B = \emptyset$, then

$$P(A|B) = 0.$$

Knowing that B occurred makes A impossible, since no outcome lies in both A and B.

Conditioning Defines a New Probability Measure

For fixed B, with $P(B) > 0$,

$$P(\cdot|B)$$

is itself a probability measure.

Thus,

$$\boxed{(\Omega, \mathcal{F}, P) \quad \longrightarrow \quad (\Omega, \mathcal{F}, P(\cdot|B)).}$$

Conditioning updates our probability model after learning that B occurred.

Bayes' Theorem

If we know $P(B|A)$, can we find $P(A|B)$?

Observe that

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A).$$

Hence,

$$\boxed{P(A|B) = \frac{P(B|A)P(A)}{P(B)}}$$

Law of Total Probability

Recall that a partition

$$A_1, \ldots, A_n$$

divides the sample space into disjoint events:

$$\Omega = A_1 \cup \cdots \cup A_n,$$

Therefore,

$$B = (B \cap A_1) \cup \cdots \cup (B \cap A_n).$$

Since these events are disjoint,

$$P(B) = \sum_{i=1}^{n} P(B \cap A_i).$$

Using

$$P(B \cap A_i) = P(B|A_i)P(A_i),$$

we obtain

$$\boxed{P(B) = \sum_{i=1}^{n} P(B|A_i)P(A_i).}$$

Bayes' Theorem with a Partition

Combining Bayes' theorem with the law of total probability,

$$\boxed{P(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum_{j=1}^{n} P(B|A_j)P(A_j)}}$$

Interpretation of Bayes' Rule

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{P(B)}.$$

$$\underbrace{P(A_i)}_{\text{Prior}} \qquad \underbrace{P(B|A_i)}_{\text{Likelihood}} \qquad \underbrace{P(A_i|B)}_{\text{Posterior}}$$

where

$$P(B) = \sum_j P(B|A_j)P(A_j)$$

is the probability of observing the evidence B.

$$\boxed{\text{Prior belief} + \text{new observation} \longrightarrow \text{updated belief.}}$$

Engineering Example: Detection

An unknown symbol $S \in \{1, \ldots, N\}$ is transmitted. At the receiver, we observe $Y \in \{1, \ldots, N\}$. The goal is to estimate (infer) the transmitted symbol. We call this estimate $\hat{S}$.

Likelihood

The channel is characterized by

$$P(Y = y|S = s).$$

For a fixed observation $Y = y$, $P(Y = y|S = s)$, viewed as a function of s, is called the **likelihood**.

It quantifies how likely the observation $Y = y$ is when symbol $S = s$ is transmitted.

Posterior probability

After observing $Y = y$, what is the probability that symbol s was transmitted?

Answer:

$$P(S = s | Y = y),$$

called the **posterior probability**.

Bayes' theorem computes the posterior from the likelihood:

$$P(S = s \,|\, Y = y) = \frac{P(Y = y \,|\, S = s)\, P(S = s)}{\sum_{i=1}^{N} P(Y = y \,|\, S = i)\, P(S = i)}.$$

Thus, posterior probabilities can be computed from

- prior probabilities,
- likelihoods,
- the law of total probability.

Maximum Likelihood (ML) Detection

Suppose the prior probabilities are unknown. Choose

$$\hat{s}_{ML} = \arg\max_{s} P(Y = y|S = s).$$

This is called a *Maximum Likelihood detector.* ML ignores prior probabilities.

Maximum A Posteriori (MAP) Detection

If the prior probabilities are available,

$$\hat{s}_{MAP} = \arg\max_{s} P(S = s | Y = y).$$

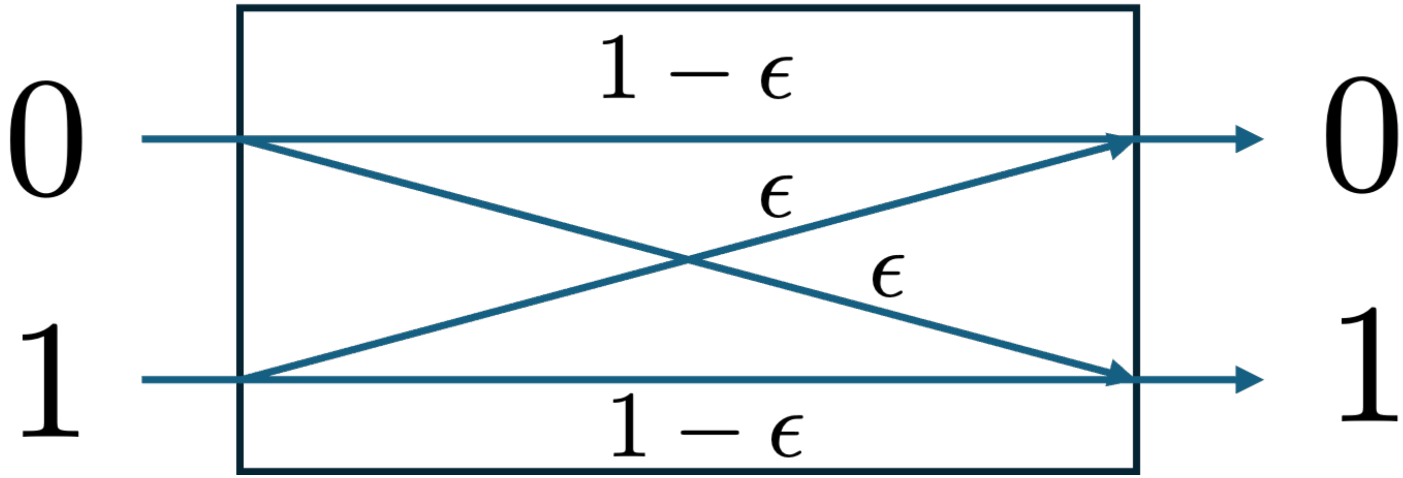
This is called a *Maximum A Posteriori* detector. MAP exploits prior information.

Using Bayes,

$$\hat{s}_{MAP} = \arg\max_{s} P(Y = y | S = s) P(S = s),$$

since the denominator does not depend on s.

Example: Binary Symmetric Channel



Source:

$$P(S=0)=0.9, \qquad P(S=1)=0.1.$$

Channel:

$$P(Y \neq s|S=s)=\varepsilon, \text{ where } \varepsilon \leq \frac{1}{2}.$$

Maximum Likelihood (ML)

Suppose $Y = 1$ is observed. The likelihoods are

$$P(Y = 1|S = 0) = \varepsilon,$$

$$P(Y = 1|S = 1) = 1 - \varepsilon.$$

Since $1 - \varepsilon \geq \varepsilon$, the ML estimator is $\hat{S}_{\mathrm{ML}}(1) = 1$. Similarly, if $Y = 0$ is observed, then $\hat{S}_{\mathrm{ML}}(0) = 0$. Overall:

$$\boxed{\hat{S}_{\mathrm{ML}}(Y) = Y.}$$

Maximum A Posteriori (MAP)

Using Bayes' theorem,

$$P(S = s|Y = 1) \propto P(Y = 1|S = s)P(S = s).$$

Hence,

$$P(S = 0|Y = 1) \propto 0.9\,\varepsilon,$$

$$P(S = 1|Y = 1) \propto 0.1(1 - \varepsilon).$$

Therefore,

$$\boxed{\hat{S}_{\text{MAP}}(1) = \begin{cases} 0, & 0.9\varepsilon \geq 0.1(1-\varepsilon), \\ 1, & 0.9\varepsilon < 0.1(1-\varepsilon). \end{cases}}$$

Similarly:

$$\boxed{\hat{S}_{\text{MAP}}(0) = \begin{cases} 0, & 0.9(1-\varepsilon) \geq 0.1\varepsilon, \\ 1, & 0.9(1-\varepsilon) < 0.1\varepsilon \end{cases} = 0.}$$

In other words, the MAP detector looks like:

if $0 \leq \varepsilon \leq 0.1$, then $\hat{S}_{\text{MAP}}(Y) = Y$;

if $0.5 \geq \varepsilon > 0.1$, then $\hat{S}_{\text{MAP}}(Y) = 0$.

Probability of Error

The probability of error is

$$P_e = P(\hat{S} \neq S).$$

For a BSC with $\varepsilon \leq \frac{1}{2}$, the ML detector is

$$\hat{S}_{\mathrm{ML}}(Y) = Y.$$

Hence,

$$P_e^{\mathrm{ML}} = P(\hat{S}_{\mathrm{ML}} \neq S) = P(Y \neq S)$$

$$= P(Y \neq S|S = 0)P(S = 0) + P(Y \neq S|S = 1)P(S = 1) = \varepsilon.$$

The MAP rule is

$$\hat{S}_{\mathrm{MAP}}(Y) = \begin{cases} 0, & \varepsilon > \frac{1}{10}, \\ Y, & \varepsilon \le \frac{1}{10}. \end{cases}$$

Therefore,

$$P_e^{\mathrm{MAP}} = \begin{cases} \varepsilon, & 0 \le \varepsilon < \frac{1}{10}, \\ 0.1, & \frac{1}{10} \le \varepsilon < \frac{1}{2}. \end{cases}$$

Remark.

$$\boxed{P_e^{\mathrm{MAP}} \leq P_e^{\mathrm{ML}}}$$

with strict inequality for

$$\frac{1}{10} < \varepsilon < \frac{1}{2}.$$

$$\boxed{\text{Prior information improves detection performance.}}$$